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Advanced Sentiment Analysis in Health Care Using Hybrid Transformer with LSTM-Based Deep Learning: A Social Media Data Perspective

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Abstract: Sentiment Analysis (SA), a sub-field of Natural Language Processing (NLP), automatically classifies sentiment in unstructured text data, offering numerous benefits in healthcare, including improved patient outcomes and enhanced quality of care. In Healthcare, which utilizes advanced technologies like machine learning, AI, and IoT for personalized healthcare, SA plays a crucial role in analyzing medical information. Twitter has become a key platform for sharing healthcare-related data, making it essential for medical research. However, existing deep learning models for sentiment analysis of social media data have many limitations. Additionally, these models require large labeled datasets and face challenges while integrating with other techniques. To address these issues, this paper presents a Hybrid Model of Transformer and LSTM (HTLSTM) for sentiment analysis model that explores syntactic and semantic information, sentimental analysis, and contextual analysis of the data. This model aims to provide a more accurate and efficient sentiment analysis framework for healthcare-related social media data. The proposed hybrid classification performance strategy yielded an accuracy of 97.89 percent and an F1-score of 98.93 percent, according to the experimental findings obtained from the WebMD drug data set.

Keywords: Health care, Transformer learning, Social media drug reviews, Sentiment analysis, LSTM.

1. Introduction

In today's evolving healthcare landscape, patients' experience and satisfaction have become paramount. Healthcare providers are continually seeking innovative strategies to enhance revenue growth and gain a competitive advantage in the market. Moreover, they are committed to fostering positive relationships with their patients, recognizing that a patient-centric approach is essential for building trust, loyalty, and long-term success. Informed decisions on drug effects can be made by identifying patterns and associations within the data [1]. Sentiment analysis in healthcare is a valuable tool for gauging patient's perceptions and experiences regarding their care and treatment [2]. By leveraging data mining techniques, healthcare providers can uncover insightful opinions, attitudes, and emotions expressed by patients through various channels. Utilizing artificial intelligence and natural language processing, sentiment analysis categorizes patient feedback into polarity, emotions, and intentions [3-5].

Sentiment Analysis in Healthcare represents a groundbreaking approach to patient experience analytics, harnessing the power of Machine Learning (ML) and Deep Learning (DL) to extract nuanced insights from patient feedback, social media, and online reviews [6, 7]. This innovative technology enables healthcare providers to move beyond traditional sentiment analysis, identifying subtle emotions and sentiment shifts, detecting early warning signs of patient dissatisfaction, uncovering hidden patterns and trends in patient feedback, providing personalized recommendations for quality improvement, and enhancing patient engagement and empowerment [8, 9].

Advanced Sentiment Analysis in Healthcare transforms patient experience, improves outcomes, and positions healthcare providers for success in the competitive healthcare landscape [10]. With its ability to analyze vast amounts of data, this

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technology provides a comprehensive understanding of patient sentiments, emotions, and intentions, enabling healthcare providers to tailor their services to meet the evolving needs of their patients, improve population health, and reduce costs. As the healthcare industry continues to evolve, Future patient-centered treatment will be greatly influenced by advanced sentiment analysis in the healthcare industry. Fig. 1 illustrates the general process of sentiment analysis.

In sentiment classification, polarity detection is a NLP method used to decide the sentiment of text data. It analyzes the language used to detect whether the sentiment is expressed as positive, Negative and Neutral.

The sentiment analysis process involves a series of systematic steps, beginning with data preprocessing, which encompasses data preprocessing to enhance pre-trained embeddings [11]. This is followed by the application of advanced DL techniques, with RNNs, LSTMs, and ERT models, to perform sentiment analysis [12-14].

Sentiment analysis (SA) on drug reviews is a crucial task for understanding patient opinions and experiences. Recent studies have leveraged deep learning advancements, including CNN and LSTM networks [15], BERT with Bi-LSTM [16], and hybrid CNN-LSTM [17] architectures, to improve sentiment analysis outcomes. However, existing solutions often struggle with accurately capturing nuanced sentiments and contextual relationships in text data. The paper addressed these margins by exploring the effectiveness of a proposed HTLSTM model in analyzing sentiment in drug reviews, with the goal of achieving improved accuracy and reliability in sentiment classification. By investigating the possible of HTLSTM, paper seeks to add to the progress of more sophisticated sentiment analysis tools for healthcare applications

The goal of the hybrid model is to optimize sentiment analysis by integrating the benefits of:

- TF-IDF (Term Frequency-Inverse Document Frequency) and correlation analysis for feature extraction and representation learning.
- Transformer for contextualized representations and capturing long-range dependencies.
- LSTM (Long Short-Term Memory) for modeling temporal dynamics and sequential relationships.

SA is a crucial task in Natural Language Processing (NLP) that involves determining the emotional tone or sentiment of text data. With the increasing amount of online data, SA has become a vital tool for understanding public opinions, preferences, and emotions. Recent studies have explored SA in various domains, including healthcare, where it can provide valuable insights into patient experiences, disease trends, and treatment outcomes. This related work summarizes recent advancements in SA, including its applications in healthcare, computational linguistics, and deep learning.

SA on drug reviews is a crucial task for understanding patient opinions and experiences. Recent studies have leveraged deep learning advancements, including CNN and LSTM networks, BERT with Bi-LSTM, and hybrid CNN-LSTM architectures, to improve sentiment analysis outcomes. However, existing solutions often struggle with accurately capturing nuanced sentiments and contextual relationships in text data. This paper addresses these limitations by exploring the effectiveness of a proposed Hybrid Transformer-LSTM (HTLSTM) model in analyzing sentiment in

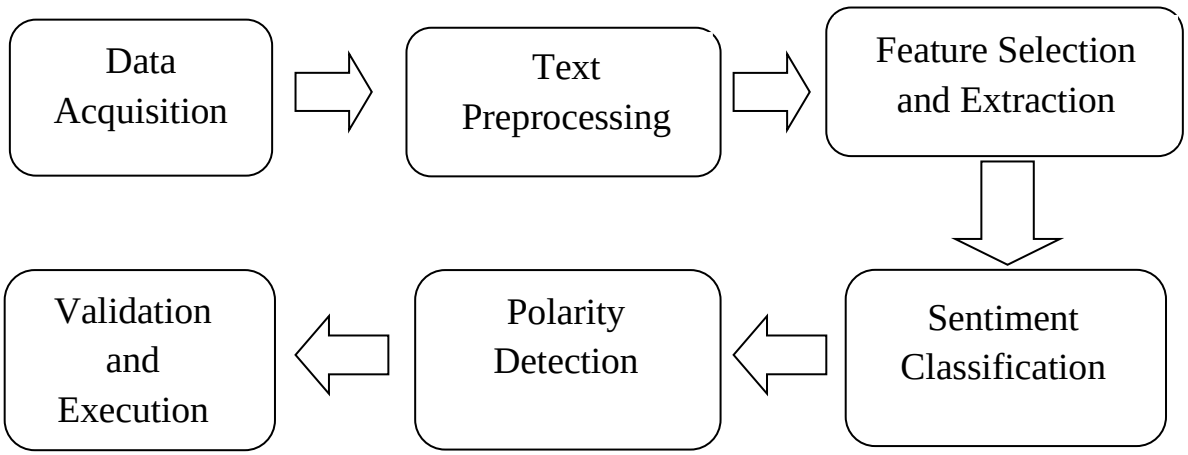

Figure. 1 A general method for sentiment analysis

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drug reviews, with the goal of achieving improved accuracy and reliability in sentiment classification. This paper seeks to contribute to the development of more sophisticated sentiment analysis tools for healthcare applications. The remainder of this paper is organized as follows: Section 2 reviews related work on sentiment analysis, Section 3 presents the proposed HTLSTM model, Section 4 discusses the experimental results, and Section 5 concludes the paper with future directions.

2. Related work

Rani and Kumar (2019) [18] explored the task of SA on natural language text, a crucial and challenging endeavor in the realm of NLP. Previous research has employed various SA techniques, including lexicon-based and machine learning approaches, for multiple languages such as English and Chinese. The author's experiments aimed to assess the effectiveness of CNN-based SA on Hindi text, contributing to the advancement of NLP research in non-English languages.

Ramírez et al. (2019) [19] researched sentiment analysis applications in healthcare. They noted that subjective information, such as opinions and attitudes, is crucial for informed decision-making. Despite its potential, sentiment analysis in healthcare has been understudied. To address this, the authors proposed a module for extracting sentiments and emotions from healthcare-related texts, leveraging natural language processing and computational linguistics techniques. This approach can uncover valuable insights into patient experiences, disease trends, and treatment outcomes, ultimately benefiting healthcare stakeholders.

Abualigah et al. (2020) [20] explored the role of computational linguistics in analyzing mental health knowledge derived from online sources, including the web and social media. Their analysis quantified global societal attitudes and sentiments towards specific entities, products, and ideas, revealing the contextual duality of knowledge. Sentiment analysis, a technique used in various approaches, including product and service reviews, has also been applied in healthcare. The vast amount of online healthcare information, including personal blogs, social media, and medical issue ratings, presents an opportunity for sentiment analysis to provide valuable insights. By leveraging medical information, sentiment analysis can enhance healthcare quality and outcomes. The authors presented a comprehensive review of sentiment analysis methods and techniques specifically applied in the medical domain, highlighting their benefits and potential applications.

Wang et al. (2021) [21] conducted a comprehensive study on the classification of unbalanced data sets, a crucial topic in machine learning. Initially, they provided an overview of unbalanced data sets and then delved into a detailed analysis of various classification methods from multiple perspectives, including data sampling, algorithmic approaches, feature-level techniques, cost-sensitive functions, and deep learning strategies. Furthermore, they categorized data sampling methods into distinct technologies, such as SMOTE, SVM, and KNN, and compared their advantages and disadvantages. Finally, they summarized evaluation criteria for unbalanced data set classifiers and outlined future research directions, providing a valuable resource for scholars and practitioners in the field.

Ghosh et al. (2023) [22] introduced a novel pain detection system that uses advanced techniques to analyze facial expressions and detect pain sentiments within a smart healthcare framework. The implementation is divided into following components: face region detection, feature analysis, pain intensity, and score fusion. This multi-component approach enables the system to accurately detect pain sentiments and intensity levels, contributing to the advancement of smart healthcare technologies.

Chen, Y (2024) [23] investigated the application of BERT in the medical domain, exploring the efficacy of various deep learning networks as output layer modules. Specifically, Chen examined the performance of CNN and Fully Connected Network (FCN) when integrated with BERT on the METS-CoV dataset.

The section highlights the significance of Sentiment Analysis in various domains, including healthcare, computational linguistics, and deep learning. The studies discussed demonstrate the effectiveness of SA in extracting valuable insights from text data, improving healthcare outcomes, and advancing smart healthcare technologies. The applications of SA range from analyzing patient experiences and disease trends to detecting pain sentiments and intensity levels. Furthermore, the integration of DL networks with BERT has shown promising outcomes in the medical domain. Overall, the related work emphasizes the importance of SA in understanding human emotions and opinions, paving the way for future research in NLP and its applications.

Table 1 provides a concise overview of various existing sentiment analysis (SA) techniques and their limitations.

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Table 1. Limitations of Existing Sentiment Analysis Techniques

References	Comparison of different existing SA methods drawbacks	
	Methods	***Drawbacks***
[18]	Lexicon-based approach	Limited domain-specific knowledge, relies on pre-defined dictionaries, may not capture nuanced sentiments
[19]	Traditional sentiment analysis	Often struggle to capture contextual relationships, may not perform well in healthcare domain
[20]	Computational linguistics approach	May not quantify global societal attitudes and sentiments accurately, limited analysis of mental health knowledge
[21]	Data sampling methods	May not always improve classification performance, can be sensitive to parameter settings
[22]	Traditional pain detection systems	May not accurately detect pain sentiments and intensity levels without advanced techniques, limited analysis of facial expressions
[23]	Deep learning methods	May not capture contextualized representations, may not perform well on small datasets, limited domain-specific knowledge

3. Method

The proposed research methodology performs the Sentiment Analysis using Hybrid Model of Transformer and LSTM (HTLSTM) model. The hybrid model combines TF-IDF and correlation analysis for feature extraction, Transformer for contextualized representations, and LSTM for temporal dynamics, to improve sentiment analysis accuracy and effectiveness, leveraging the strengths of each technique to generate robust and contextualized features for sentiment classification procedure is discussed in this section. The overall proposed process flow diagram is illustrated in Fig. 2.

3.1 Data collection

Most existing research on drug reviews from online forums and Twitter has primarily focused on sentiment analysis, leaving a significant gap in research on classifying drugs based on their efficacy.

This study aims to address this gap by classifying drugs according to their effectiveness, using data extracted from public online forums like (WebMD.com) [24]. These forums contain reviews from patients who have used drugs for pain relief, providing valuable insights into their experiences.

This paper focuses on three critical features: condition, review, and satisfaction, which are deemed significantly influential in the analysis. A thorough examination and analysis of these three features are crucial, as they collectively provide essential information for shaping the study's insights and outcomes. The extracted data includes review comments and ratings provided by patients, stored in a CSV format is shown in Fig. 3.

3.2 Data cleaning or data pre-processing

Data preprocessing transforms raw data into an analysis-ready state by resolving inconsistencies, handling missing values, removing outliers, and correcting errors, laying the groundwork for reliable

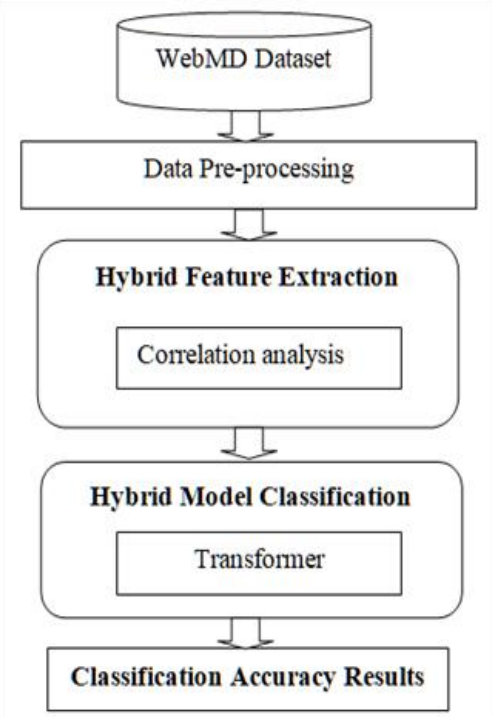

Figure. 2 Proposed Flow Diagram

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	A	B
1	**Reviews**	**Effectiveness**
2	I'm a retired physician and of all the meds I have tried for my allergies (seas	5
3	cleared me right up even with my throat hurting it went away after taking th	5
4	why did my PTINR go from a normal of 2.5 to over 100?	3
5	FALLING AND DON'T REALISE IT	2
6	My grandfather was prescribed this medication (Coumadin) to assist in bloc	1
7	help heart condition operation well	4
8	Haven't gotten pregnant so it does it's job. I was switched to this brand fror	5
9	I have take this for 5 years age 45-50 to prevent ovarian cysts (I had 2 burs	5
10		2
11	The 12 hour spray only works for me for 6 hours.	2
12	Excellent in reducing inlamation associated with osteo-arthritis!	5
13	Taking Lyza made me break out HORRIBLY. I tried it out for 3 months and th	5
14	This stuff needs to be removed from the market! Itâ€™s so harmful and ages you fast!! I have dark racoon eyes, and dark marks all over my neck. Shortness of breath Horrible mood swings Horrible acne Enlarged poors And hair loss. Avoid this birth control because it literally IS THE PLAGUE!	1
15	I usually have zero to little acne, a few months after taking this medication r	3

Figure. 3 Input Drug Dataset

insights.However, raw data extracted from websites is often unstructured and noisy, making it imperative to preprocess and transform the data into a suitable format for analysis. This involves removing irrelevant information, correcting errors, and organizing the data into a structured format that can be easily interpreted by the algorithms, enabling accurate insights and analysis.

Data pre-processing was performed to prepare the dataset for analysis. The first step was to address missing values by removing rows with empty review comments. Subsequently, text processing was conducted to transform unstructured text into a suitable format for algorithmic analysis. To remove reviews less than a certain threshold, we need to tokenize the reviews and compute the length of each review in terms of the number of tokens. This involved a series of standard steps, as illustrated in Fig. 4, to enable the algorithms to comprehend the text data effectively.

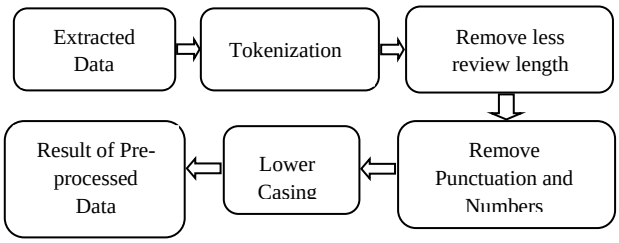

Figure. 4 Data pre-processing

3.3 Hybrid feature extraction

The hybrid approach combines the strengths of TF-IDF and correlation analysis to select features for sentiment analysis on the WebMD and DUSE datasets. First, TF-IDF scores are calculated for each feature (word) in the dataset, capturing the importance of each word in the context of the entire corpus. Next, correlation analysis is performed to measure the relationship between each feature and the sentiment label (Positive or Negative or Neutral). The correlation values are then combined with the TF-IDF scores using a weighted sum, where the weights (α and β) control the importance of each component. The resulting hybrid scores reflect both the relevance of each word to the sentiment analysis task and its correlation with the sentiment label. By extracting features with high hybrid scores, the approach identifies words that are not only informative but also highly correlated with the sentiment label, leading to improved sentiment analysis performance. The weights can be adjusted to optimize the performance of the hybrid model, allowing for flexibility in balancing the contributions of TF-IDF and correlation analysis.

Let, X be a feature (word) in the WebMD dataset; y be the sentiment label; TF-IDF(X) be the TF-IDF score for feature X; Corr(x, y) be the correlation amongattribute$X$ and sentiment label y; α be the weight for TF-IDF ($0 \leq \alpha \leq 1$); β be the weight for correlation ($0 \leq \beta \leq 1$).

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The mathematical formula for the hybrid approach is given in Eq. (1):

$$HS(x)=\alpha\times TFIDF(X)+\beta\times Corr(X,y) \quad (1)$$

Where, HS(x) is the hybrid score for feature *X*; TF-IDF(*x*) is the TF-IDF score for feature *X*; Corr(X, y) is the correlation among feature x and sentiment label *y;*α and β are weights that control the importance of TF-IDF and correlation, respectively.

3.4 Hybrid model of transformer with LSTM for sentiment analysis

The hybrid approach integrates Transformer and LSTM models for SA on the WebMD dataset, harnessing the Transformer's contextual understanding and the LSTM's ability to capture sequential patterns. The Transformer Encoder is a neural network architecture that processes input sequences of tokenized text data, such as reviews from the WebMD dataset, and generates contextualized representations of the input data.

The encoder consists of identical layers with multi-head self-attention mechanisms and position-wise feed-forward networks, utilizing layer normalization and residual connections to capture complex contextual relationships. The output is a sequence of vectors, each representing a contextualized embedding of the corresponding input token, which can be used for sentiment analysis tasks. The proposed method begins with a Transformer Encoder of BERT pre-trained model, which generates contextualized representations of the input feature selection data.

The output from the LSTM layer is then fed into a sentiment classification layer, which produces a sentiment label (Positive or Negative or Neutral).

This hybrid approach harnesses the Transformer's contextual understanding and the LSTM's sequential processing capabilities to effectively analyze sentiment in the WebMD dataset, leading to improved performance and accuracy in sentiment analysis tasks. The proposed architecture diagram illustrated in Fig. 5.

$$Tr(HS)=Encoder(HS) \quad (2)$$

$$Encoder(HS)=Dropout(MHA(HS)+PFF(HS)) \quad (3)$$

Where, *Tr* is Transformer, MHA represents Multi Head Attention; *PFF* represents Position wise Feed Forward; *X* be the input text preprocessed data (WebMD dataset).

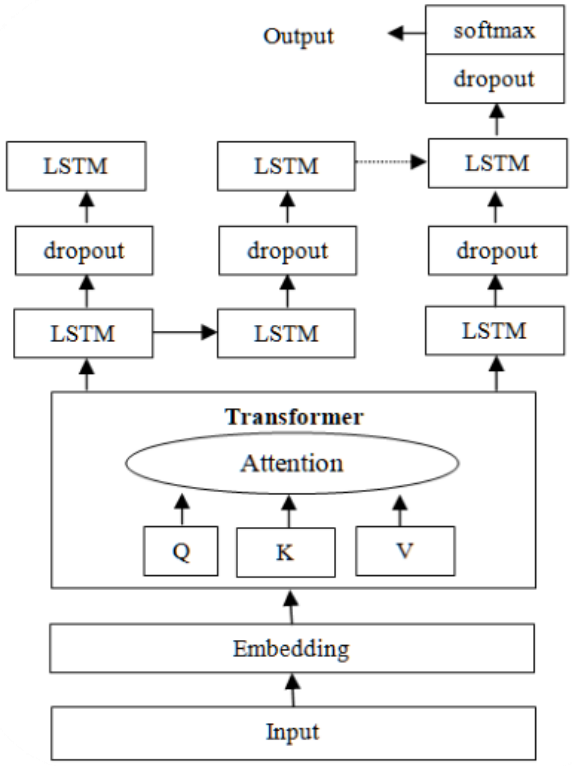

Figure. 5 Hybrid Transformer-LSTM architecture

$$MHA(HS)=Concat(h_1,h_{2,\ldots,}h_n)\cdot W^0 \quad (4)$$

Where, h_i is the outcome of the ith attention head:

$$h_i=Attention(QW_i^Q,KW_i^K,VW_i^V) \quad (5)$$

Where, *Q*, *K*, *V* are the query, key, and value matrices:

$$Q=HS\cdot W^Q,\ K=HS\cdot W^K, V=HS\cdot W^V \quad (6)$$

PFF(HS) is the position-wise feed-forward network:

$$PFF(HS)=ReLU(HS\cdot W^F+b^F) \quad (7)$$

Where *b* represents bias.

Let L(Tr(HS)) is the output of the LSTM layer defined as,

$$L(Tr(HS))=Dropout(LSTM(Tr(HS))) \quad (8)$$

$$LSTM(Tr(HS))=CS(Tr(HS))\times HdS(Tr(HS)) \quad (9)$$

Where, *CS* represents CellState; *Hds* represents HiddenState.

CS(*Tr*(*HS*)) is defined as

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$$\mathrm{CS}(\mathrm{Tr}(\mathrm{HS}))=\mathrm{f}\times\mathrm{CS}(\mathrm{Tr}(\mathrm{HS\text{-}1}))+\mathrm{i}\times\mathrm{tannh}(\mathrm{W_c}\cdot\mathrm{Tr}(\mathrm{HS})\ \mathrm{E}+\mathrm{b_c} \quad (10)$$

HdS(*Tr*(*HS*)) is defined as,

$$\mathrm{HdS}(\mathrm{Tr}(\mathrm{HS}))=\mathrm{og}\times\tanh(\mathrm{CS}(\mathrm{Tr}(\mathrm{HS}))) \quad (11)$$

Where, *og* is the output gate; *CS*(*Tr*(*HS*)) is the cell state of the LSTM layer.

The output of the Hybrid Transformer with LSTM layer is then passed through a sentiment classification layer,

$$\mathrm{Sentiment}(\mathrm{L}(\mathrm{Tr}(\mathrm{HS}))=\mathrm{softmax}(\mathrm{LinearL}(\mathrm{Tr}(\mathrm{HS}))) \quad (12)$$

Where, Linear((*L*(*Tr*(*HS*))) is a linear transformation of the outcome from the LSTM layer; softmax is the softmax activation function.

The final outcome is a likelihood division over the possible sentiment labels (positive or negative or neutral):

$$\mathrm{y}=\mathrm{argmax}(\mathrm{Dropout}(\mathrm{Sentiment}(\mathrm{L}(\mathrm{Tr}(\mathrm{HS})))) \quad (13)$$

This mathematical model combines the strengths of hybrid model for sentiment analysis on the WebMD dataset. Dropout randomly sets a fraction of the output elements to zero during training, which helps prevent over-fitting by reducing the model's reliance on any single element. The dropout rate can be tuned as a hyper-parameter to achieve the best results. The Transformer encoder captures contextual relationships in the input hybrid feature vectors, while the LSTM layer models temporal dependencies. The sentiment classification layer then uses the output from the LSTM layer to predict the sentiment label.

3.5 Sentiment classification

In algorithm 1, Sentiment Classification in Healthcare model uses a hybrid approach, combining a Transformer Encoder, LSTM layer, and dense output layer, to classify sentiment in text data. The input text undergoes preprocessing with tokenization, stopword removal, and embeddings (e.g. Word2Vec, GloVe), followed by hybrid feature selection using TF-IDF scores and correlation analysis to identify the most relevant features. The Transformer Encoder captures contextual relationships using self-attention mechanisms and positional encoding, while the LSTM layer captures sequential dependencies and temporal relationships. Finally, the output is passed through a dense layer with softmax activation function to predict sentiment labels, and the model is trained on labeled text data using a proper optimizer and loss function to enable prediction of sentiment in new, unseen text data.

Table 2 provides the Notation List for Hybrid Transformer-LSTM Model.

Algorithm 1: Hybrid Transformer with LSTM for sentiment analysis

Input: WebMD dataset *(reviews, sentiment labels)*
Output: Sentiment result
Preparation:
a) Data Collection
b) Data Preprocessing
c) Hybrid Feature selection using TF-IDF with Correlation
d) Sentiment prediction using Hybrid Model of Transformer and LSTM method
e) Compute Accuracy and Loss

Methods:
1. *Data preprocessing*
2. *Hybrid feature selection*
   - Calculate TF-IDF scores for each review in the text data.
   - Perform correlation analysis on the TF-IDF scores to identify the most relevant features.
   - Select the top *N* features based on the correlation analysis.
3. *Hybrid Transformer-LSTM Model*
   - Use the selected features from Step 2 as input to the Transformer encoder (BERT pre-trained model).
   - Pass the output of the Transformer encoder with the LSTM layer.
   - Use the output of the LSTM layer as key to the sentiment classification layer.
4. *Sentiment Classification*
   - Train the hybrid Transformer-LSTM model on the text data using the sentiment label.
   - Predict sentiment of new reviews using the trained model.

4. Discussion

The paper presents a sentiment analysis of the WebMD and DUSE datasets, focusing on extracting insights from online health-related reviews and discussions to understand public sentiments and opinions about various health topics, treatments, and

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Table 2. Proposed notation list

Symbol	Notations
α	Weight for TF-IDF ($0 \leq \alpha \leq 1$)
B	Weight for correlation ($0 \leq \beta \leq 1$)
Tr	Transformer
Q	Query Matrix
K	Key Matrix
V	Value Matrix
PFF	Position-wise Feed-Forward network
f	Forget gate
Og	Output gate
CS	Cell state
y	Sentiment label

services, potentially leveraging natural language processing techniques to classify sentiments. For result estimation, the Hybrid Transformer-LSTM Model (HTLSTM) approach was utilized. Using Python 3.8 simulations on Windows 10 with 8GB of main memory and Intel Core i3-10100F series 4.30 GHz based CPUs, the results are achieved. Specifically the proportions used for training, testing, and validation sets (e.g., 80% for training, 10% for validation, and 10% for testing).

From Figs. 6 and 7 illustrate the training accuracy and loss results of the proposed HTLSTM method, demonstrating its performance over time. The accuracy plot shows a steady increase, indicating effective learning, while the loss plot exhibits a corresponding decrease, signifying convergence and minimal error.

From Fig. 8, a word cloud for sentiment analysis, visually representing the most prominent words in the WebMD dataset, with word size and prominence indicating frequency and importance, providing insight into the dominant themes, emotions, and

Figure. 6 Proposed Hybrid Model Training and Validation Accuracy

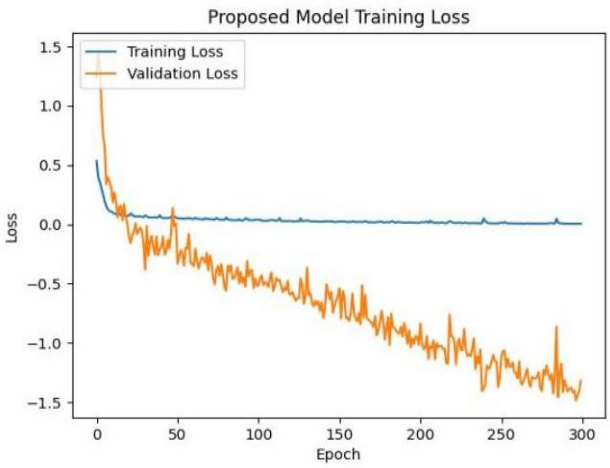

Figure. 7 Proposed Hybrid Model Training and Validation Loss

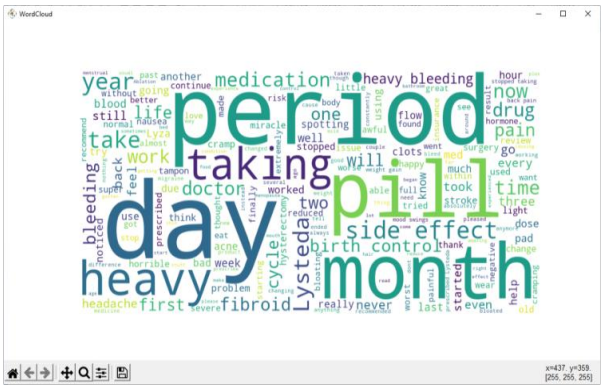

Figure. 8 WordCloud result

sentiments expressed in the text data.

From Fig. 9, the sentiment analysis polarity score that determines the sentiment of a patient reviews. If the polarity score is greater than 0, it indicates a positively polarized comment, while a score less than 0 indicates a negatively polarized comment, and a score equal to 0 indicates a neutrally polarized comment, allowing for a straightforward classification of sentiment.

Tables 3 to 6 presents a comparative analysis of the classification performance of existing models, SVM, KNN, RF [25], CNN, LSTM, Bi-LSTM [26] and Bi-LSTM-CNN [27], with the proposed HTLSTM model, showcasing metrics such as Precision, Recall, Accuracy, and F1-score, highlighting the superior performance of the HTLSTM model.

From Fig. 10 presents a analysis of precision metrics across various ML models, including traditional models like SVM (87.15), KNN (88.75), and RF (89.77), as well as DL models such as CNN (90.5), LSTM (91.43), Bi-LSTM (92.3), and

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                                        Reviews  Polarity
I'm a retired physician and of all the meds I ...  0.337500
cleared me right up even with my throat hurtin...  0.285714
why did my PTINR go from a normal of 2.5 to ov...  0.150000
                          FALLING AND DON'T REALISE IT  0.000000
My grandfather was prescribed this medication ...  0.046429
                                                ...        ...
my 44 year old sister is in the neuro ICU from...  0.077143
Heavy periods all my life. My doc found I had ...  0.035154
Thank you Ferring pharmaceuticals! I have trie...  0.039537
Was given lysteda to help with heavy periods a... -0.075000
I have been menstruating for two years straigh...  0.075000
```

Figure. 9 Sentiment Polarity result

Table 3. Precision metrics are compared to the current and suggested models

Models	Precision
SVM	87.15
KNN	88.75
RF	89.77
CNN	90.5
LSTM	91.43
Bi-LSTM	92.3
BiLSTM-CNN	94.29
Proposed HTLSTM	**99.52**

Table 4. Recall metrics are compared to the current and suggested models

Models	Recall
SVM	65.97
KNN	69.27
RF	71.5
CNN	73.11
LSTM	75.3
Bi-LSTM	77.42
BiLSTM-CNN	89.10
Proposed HTLSTM	**98.34**

Table 5. Accuracy metrics are compared to the current and suggested models

Models	Accuracy
SVM	64.80
KNN	67.60
RF	69.55
CNN	71
LSTM	73
Bi-LSTM	75
BiLSTM-CNN	87
Proposed HTLSTM	97.89

BiLSTM-CNN (94.29), with the proposed HTLSTM model demonstrating superior performance with a precision of 99.52, indicating its potential for highly accurate predictions and classification result.

From Fig. 11 presents a analysis of recall metrics

Table 6. F1-score metrics are compared to the current and suggested models

Models	F1-score
SVM	75.1
KNN	77.81
RF	79.6
CNN	80.88
LSTM	82.58
Bi-LSTM	84.21
BiLSTM-CNN	92.71
Proposed HTLSTM	**98.93**

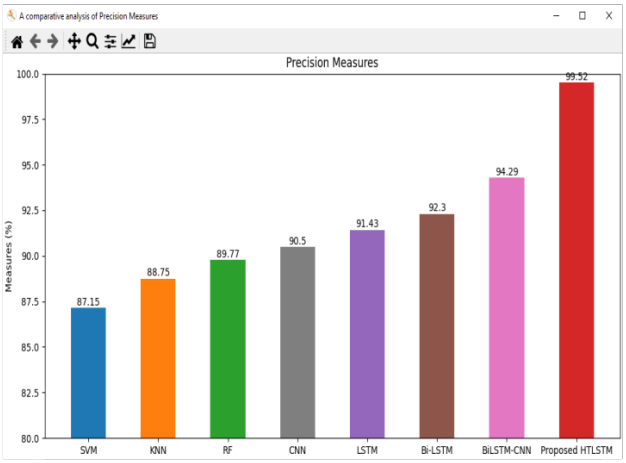

Figure. 10 Precision measure result

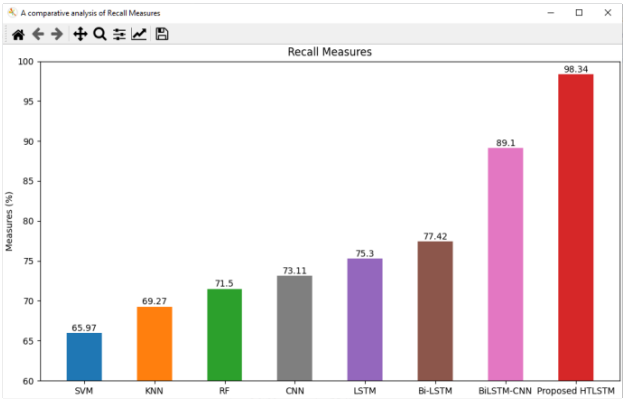

Figure. 11 Recall measure result

across various ML models, showcasing the performance of traditional models like SVM (65.97), KNN (69.27), and RF (71.5), alongside deep learning models such as CNN (73.11), LSTM (75.3), Bi-LSTM (77.42), and BiLSTM-CNN (89.10), with the proposed HTLSTM model achieving the maximum recall of 98.34, demonstrating its excellent ability to properly identify true positives and weight the efficiency of its architecture in relevant models in the WebMD data.

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From the Fig. 12 presents a analysis of accuracy metrics across various ML models, including SVM (64.80), KNN (67.6), RF (69.55), CNN (71), LSTM (73), Bi-LSTM (75), and BiLSTM-CNN (87), with the HTLSTM method demonstrating better results with an accuracy of 97.89, indicating its exceptional capability in making correct predictions and classifications, and underscoring the effectiveness of its architecture in WebMD data.

From the Fig. 13 presents a comparative analysis of F1-score metrics across various machine learning models applied to the WebMD dataset, showcasing the performance of SVM (75.1), KNN (77.81), RF (79.6), CNN (80.88), LSTM (82.58), Bi-LSTM (84.21), and BiLSTM-CNN (92.71), with the proposed HTLSTM model achieving the highest F1-score of 98.93, demonstrating its exceptional balance between precision and recall in analyzing patient reviews and sentiment on WebMD, and highlighting its potential for highly accurate and reliable predictions in healthcare-related text analysis.

The comparative analysis of various ML models on the WebMD dataset across four key performance metrics - Precision, Recall, Accuracy, and F1-score - reveals that the proposed HTLSTM model consistently outperforms existing models, achieving exceptional scores of 99.52 in Precision, 98.34 in Recall, 97.89 in Accuracy, and 98.93 in F1-score, demonstrating its superior capability in analyzing patient reviews and sentiment on WebMD, and highlighting its potential for reliable and accurate predictions in healthcare-related text analysis.

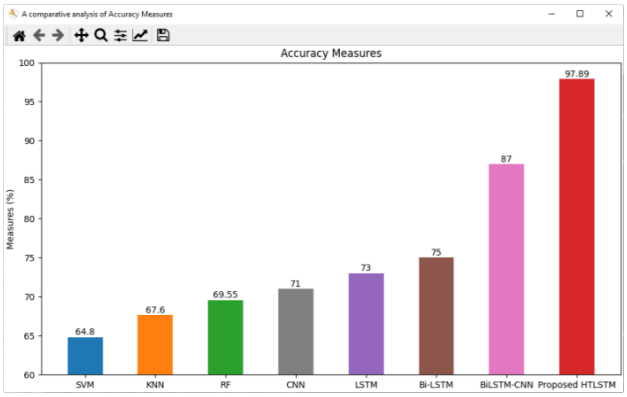

Figure. 12 Accuracy measure result

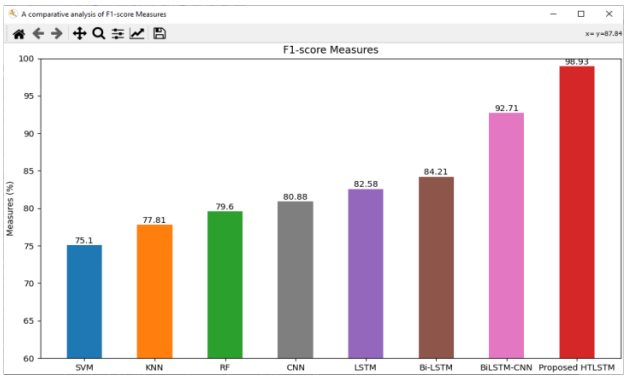

Figure. 13 F1-score measure result

5. Conclusion

This article presents a HTLSTM model combines the strengths of both techniques, capturing contextualized representations and temporal dynamics to improve sentiment classification. The proposed method begins with a Transformer Encoder of BERT pre-trained model, which generates contextualized representations of the input feature selection data, followed by an LSTM layer that processes the sequence of vectors and captures temporal dependencies. Moreover, it can be concluded that using Hybrid feature selection with HTLSTM method major progress in the performance of sentiment classification. Future work includes experimenting with pre-trained Transformers and LSTM variants, incorporating attention mechanisms, evaluating on diverse datasets, and exploring transfer learning and domain adaptation to improve generalizability.The results outperformed that the HTLSTM model attained the better accuracy of 97.89 percent for SA of patients' drug reviews (WebMD dataset) with Hybrid feature selection technique.

Conflicts of interest

The authors declare no conflict of interest.

Author contributions

Conceptualization, KTM and EIS; methodology, KTM and EIS; software, KTM; validation, KTM and EIS; formal analysis, KTM; investigation, EIS; resources, KTM; data curation, IES; writing—original draft preparation, KTM; writing—review and editing, IES; visualization, KTM; supervision, IES; project administration, IES.

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